

Modeling of Excitations in Vacancy-Ordered Double Perovskites

Efrain Martinez

July 2026

1 Introduction

Vacancy-ordered double perovskites are chemically tunable and lead-free semiconductors with the formula A_2BX_6 . Their structures can be viewed as spatially separated BX_6 octahedra, with the spaces between them occupied by A -site cations. The weak connectivity between neighboring octahedra can produce unusual excited state behavior including excitations that may be localized near a single octahedron or extend over a larger portion of the crystal.

To investigate the influence of the local environment on these excitations, we calculated excited states for two different cluster models. The first model, M0, contains only an isolated BX_6^{2-} octahedron. The second model, M1, contains the same octahedron together with eight explicit neighboring Cs atoms and a surrounding point charge environment that neutralizes the embedded model. These models allow us to test how strongly the calculated excitations depend on the neighboring Cs atoms and surrounding environment.

We studied six Cs_2BX_6 materials with $B = \text{Sn, Ti, Zr}$ and $X = \text{Br, I}$, resulting in twelve total M0 and M1 model calculations. These materials are expected to have different electronic and excited-state behavior because of the different metal and halide atoms.

2 Computational Methods

To create the cluster models we first relaxed the periodic structures using VASP with the PBEsol exchange–correlation functional. The calculations used a plane wave cutoff of 520 eV and a $5 \times 5 \times 5$ Gamma-centered k -point. These relaxed structures were then used as the references for making the M0 and M1 models we use.

All of the calculations for the M0 and M1 models were done with PySCF. Ground state references were calculated with density-fitted restricted Hartree–Fock and the def2-SVP basis set. Effective core potentials were used for Cs, Zr, Sn, and I and the M1 calculations additionally includes the point charge potentials.

The five lowest singlet excited states were calculated using the Tamm–Dancoff approximation. Excitation energies, oscillator strengths, dominant orbital transitions, and selected natural transition orbitals were calculated where we use NTOs to provide a compact hole–electron description of the excited states.

3 Results and Discussion

All twelve M0 and M1 calculations produced the five converged singlet excited states. The isolated M0 models had first excitation energies between 3.37 and 6.24 eV while the M1 models had smaller first excitation energies between 0.031 and 0.375 eV. We plot these on a histogram in Figure 1 to compare the first excitation energies for the two models.

The large differences between the M0 and M1 excitation energies show that the explicit Cs atoms and surrounding environment can strongly influence the calculated energy spectrum. These differences shouldn't be interpreted as energy shifts of the same physical exciton though as M0 and M1 contain completely different environments and orbitals used in their calculations, so their lowest states may also have different electronic character.

We also found that the lowest state was not always the most optically important state. For four of the six M1 materials the largest oscillator strength among the first five states occurred for a higher excited state.

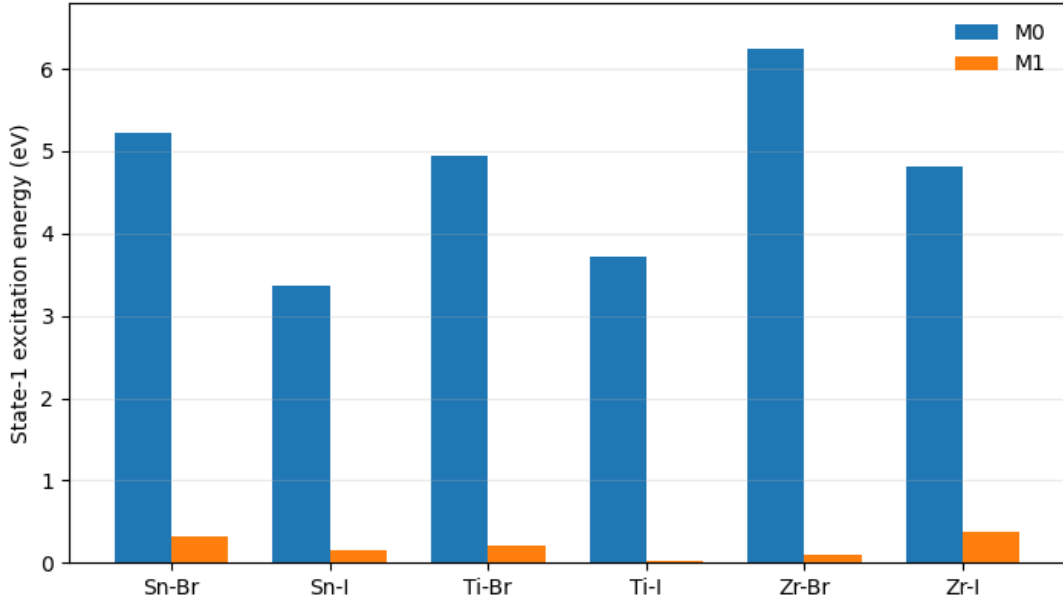


Figure 1: First singlet excitation energies calculated for the M0 and M1 models.

Figure 2 compares the first excitation energy with the energy of the maximum oscillator strength state among the five calculated M1 states. As an example, we can see Cs_2TiI_6 had a first excitation at 0.031 eV with an oscillator strength of only 1.47×10^{-4} , while its maximum-oscillator-strength state occurred at 0.420 eV. Cs_2ZrI_6 had the largest M1 oscillator strength 0.193 at its second excited state. These results show that the lowest numerical root must be distinguished from the state that is most optically important within the calculated energy range.

Most M1 first states had dominant NTO pair weights between about 0.77 and 0.98, showing that they were mainly described by one hole–electron orbital pair. However, the weights alone do not tell us for sure about their spatial localization. The present results therefore show strong sensitivity to the local model environment, but they cannot directly determine whether the excitations are localized throughout the periodic crystal.

4 Limitations and Continuation Work

The models used in this project contain only one BX_6 octahedron and thus cannot describe excitation transfer between neighboring octahedra. A larger two octahedron M2 model was constructed, but its calculations were not completed because of computational cost and convergence difficulties. With additional time, this model could be used to investigate whether the excited electron and hole extends between neighboring octahedras.

Other future work could include larger or periodic models, spatial visualization and analysis of the NTOs, diffuse or larger basis sets, and comparison with TDA-DFT or correlated excited state methods. Spin–orbit coupling may also be important, particularly for the iodine-containing compounds because iodine is a heavy element and may experience stronger relativistic effects.

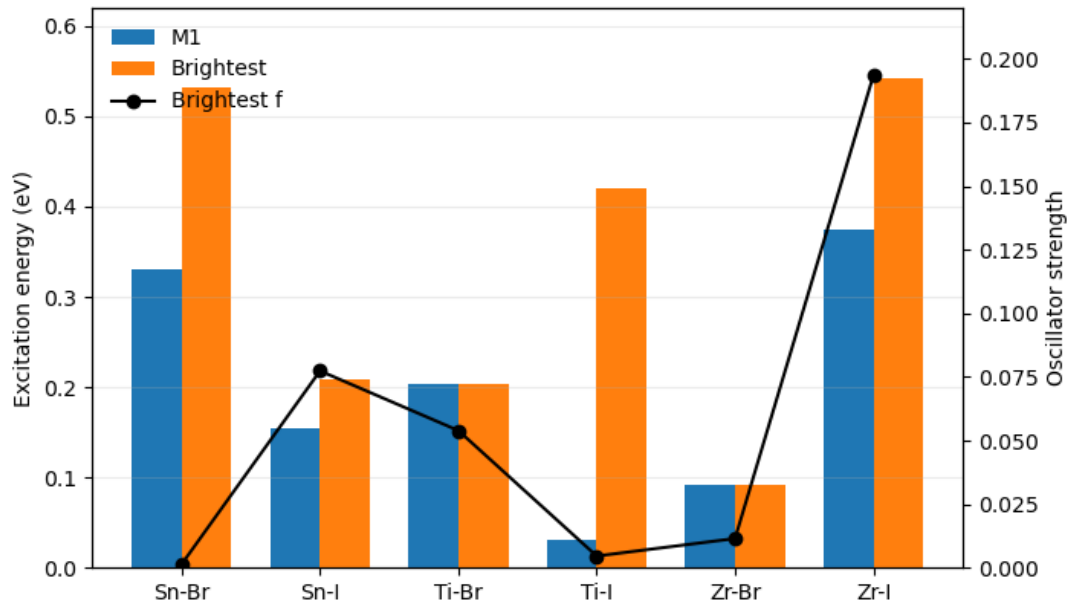


Figure 2: Comparison of the first M1 excitation and the maximum oscillator strength state among the first five. The black line and right axis show the corresponding maximum oscillator strength.